Yongho Shin

Postdoctoral researcher
Institute of Computer Science, University of Wrocław
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Research Interests

Online/approximation/learning-augmented algorithms for combinatorial optimization problems

Education

Ph.D. in Computer Science, Yonsei University, South Korea Mar. 2018 – Aug. 2024

- ◇ Dissertation topic: Relaxing hard requirements of online optimization via learning augmentation and limited revocability
- ◇ Advisor: Hyung-Chan An

B.S. in Computer Science, Yonsei University, South Korea Mar. 2012 – Feb. 2018

- ◇ Awarded *high honors at graduation*

Employment

Institute of Computer Science, University of Wrocław, Poland Nov. 2024 – present

- ◇ Postdoctoral researcher
- ◇ Advisor: Jarosław Byrka

Research Papers

Published Work

[1] Changyeol Lee, Dahoon Lee, Jongseo Lee, **Yongho Shin**, and Changki Yun. Optimal learning-augmented algorithm for online bidding. *arXiv preprint arXiv:2605.07349*, 2026. **To appear in NeurIPS 2026.**

[2] Fateme Abbasi, Martin Böhm, Jarosław Byrka, Matin Mohammadi, and **Yongho Shin**. Servicing matched client pairs with facilities. *arXiv preprint arXiv:2602.19680*, 2026. **Appeared in WAOA 2026.**

[3] Jarosław Byrka and **Yongho Shin**. Online rounding schemes for edge cover. *arXiv preprint arXiv:2507.13159*, 2025. **Appeared in WAOA 2026.**

[4] Fateme Abbasi, Hyung-Chan An, Jarosław Byrka, Changyeol Lee, and **Yongho Shin**. Approximating (weighted) chromatic correlation clustering via cluster LP. *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2026)*, 13:1-13:20, 2026.

[5] **Yongho Shin** and Phanu Vajanopath. Parsimonious learning-augmented online metric matching. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, PMLR 306, 112669–112688, 2026.

[6] Davin Choo, Billy Jin, and **Yongho Shin**. Learning-augmented online bipartite fractional matching. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 7671–7686, 2025.

[7] Changyeol Lee, **Yongho Shin**, and Hyung-Chan An. Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 70823–70855, 2025.

[8] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. *ACM Transactions on Algorithms 22(1)*, 11:1–31, 2025. *Conference version in ICML 2023 [9].*

[9] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202, 31539–31561, 2023.

[10] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for parity-constrained facility location and k-center. *Algorithmica 85*, 1883–1911, 2023. *Conference version in ISAAC 2020 [12].*

[11] **Yongho Shin** and Hyung-Chan An. Making three out of two: Three-way online correlated selection. In *Proceedings of the 32nd International Symposium on Algorithms and Computation (ISAAC 2021)*, 49:1-49:17, 2021.

[12] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for the parity-constrained facility location problem. In *Proceedings of the 31st International Symposium on Algorithms and Computation (ISAAC 2020)*, 21:1-21:17, 2020.

Preprints

[13] Jarosław Byrka, Yuhao Guo, Yang Hu, Shi Li, **Yongho Shin**, Phanu Vajanopath, Chengzhang Wan, and Zaixuan Wang. Iterative randomized rounding for better-than-2 k-median approximation. *arXiv preprint arXiv:2610.31229*, 2026.

[14] Mateusz Basiak, Marcin Bieńkowski, **Yongho Shin**, and Agnieszka Tatarczuk. Pure tail constraints for online problems. *arXiv preprint arXiv:2609.35337*, 2026.

[15] **Yongho Shin**, Changyeol Lee, and Hyung-Chan An. On optimal consistency-robustness trade-off for learning-augmented multi-option ski rental. *arXiv preprint arXiv:2312.02547*, 2023. *Merged to [8].*

[16] **Yongho Shin**, Kangsan Kim, Seungmin Lee, and Hyung-Chan An. Online graph matching problem with a worst-case reassignment budget. *arXiv preprint arXiv:2003.05175*, 2020.

Teaching

University of Wrocław, Poland

◇ Algorithms with predictions — Summer 2026

Teaching assistant, Yonsei University, South Korea

◇ CSI2103/CCO2103 Data structures — Spring 2018 – 2021, 2023, 2024

◇ CSI3108 Algorithm analysis — Fall 2018 – 2021, 2023

◇ AIC2130 Computer algorithms for AI applications — Fall 2023

◇ GEK6205 Design and analysis of optimization algorithms — Fall 2023

Awards

High honors at graduation, Yonsei University — Feb. 2018

Academic Service

- ◇ Reviewer, ICLR 2027
- ◇ Reviewer, Transactions on Machine Learning Research, 2026
- ◇ Reviewer, WAOA 2026
- ◇ Reviewer, Discrete Optimization, 2026
- ◇ Reviewer, NeurIPS 2026
  - Top reviewer
- ◇ Reviewer, MFCS 2026
- ◇ Reviewer, IWOCA 2026
- ◇ Reviewer, ICML 2026
  - Gold reviewer (top 25%)
- ◇ Reviewer, Acta Informatica, 2026
- ◇ Program committee member, EC 2026
- ◇ Reviewer, STOC 2026
- ◇ Reviewer, APPROX 2025
- ◇ Reviewer, ICALP 2025
- ◇ Reviewer, Mathematical Programming, 2024
- ◇ Reviewer, WAOA 2024

Talks and Presentations

COG Seminar, University of Wrocław, Wrocław, Poland — Oct. 2025
- ◇ Title: Optimal learning-augmented algorithm for online bidding

New Frontiers in Learning-Augmented Algorithms, Berlin, Germany — Aug. 2026
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Contributed talk

APPROX 2026, Boston, MA, USA — Aug. 2026
- ◇ Title: Approximating (weighted) chromatic correlation clustering via cluster LP

ALPS@IJCAI, Bremen, Germany — Aug. 2026
- ◇ Title: Optimal learning-augmented algorithm for online bidding
- ◇ Contributed talk

Combinatorial Algorithms Seminar, Inha University, Incheon, South Korea — July 2026
- ◇ Title: Primal-dual algorithms for edge-colored clustering

ICML 2026, Seoul, South Korea — July 2026
- ◇ Title: Parsimonious learning-augmented online metric matching
- ◇ Poster presentation

HALG 2026, Stockholm, Sweden — June 2026
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Contributed talk and poster presentation

SNU Math Seminar, SNU, Seoul, South Korea — Jan. 2026
- ◇ Title: Online rounding for set cover under subset arrivals

Combinatorics Seminar, Hanyang University, Seoul, South Korea — Jan. 2026
- ◇ Title: Primal-dual algorithms for edge-colored clustering

SNU CSE Seminar, SNU, Seoul, South Korea — Dec. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

NeurIPS 2025, San Diego, CA, USA — Dec. 2025
- ◇ Title: Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach
- ◇ Poster presentation

NeurIPS 2025, San Diego, CA, USA — Dec. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Poster presentation

2025 Combinatorics Workshop, IBS DIMAG, Daejeon, South Korea — Aug. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching
- ◇ Contributed talk

DGIST BRL AGSTA Seminar, DGIST, Daegu, South Korea — Aug. 2025
- ◇ Title: Online rounding for set cover under subset arrivals

Discrete Analysis Seminar, Yonsei University, Seoul, South Korea — Aug. 2025
- ◇ Title: Online rounding for set cover under subset arrivals

Combinatorial Optimization Seminar, Yonsei University, Seoul, South Korea — Aug. 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

HALG 2025, Zurich, Switzerland — June 2025
- ◇ Title: Learning-augmented algorithms for the multi-option ski rental problem
- ◇ Contributed talk and poster presentation

COG Seminar, University of Wrocław, Wrocław, Poland — May 2025
- ◇ Title: Learning-augmented online bipartite fractional matching

COG Seminar, University of Wrocław, Wrocław, Poland — Dec. 2024
- ◇ Title: Online correlated selection

Discrete Analysis Seminar, Yonsei University, Seoul, South Korea — June 2024
- ◇ Title: Three-way online correlated selection

Discrete Math Seminar, IBS DIMAG, Daejeon, South Korea — May 2024
- ◇ Title: Three-way online correlated selection

ICML 2023, Honolulu, HI, USA — July 2023
- ◇ Title: Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis
- ◇ Poster presentation

Theory Tea, Cornell University, Ithaca, NY, USA — Dec. 2022
- ◇ Title: Three-way online correlated selection

ISAAC 2021, Fukuoka, Japan (virtual) — Dec. 2021
- ◇ Title: Making three out of two: Three-way online correlated selection

AAAC 2021, Tainan, Taiwan (virtual) Oct. 2021

◇ Title: Making three out of two: Three-way online correlated selection

ISAAC 2020, Hong Kong, China (virtual) Dec. 2020

◇ Title: Constant-factor approximation algorithms for the parity-constrained facility location problem

Experience

Co-organizer of Yonsei CS theory student group, Yonsei University, South Korea Jan. 2023 – Feb. 2024

◇ Initiated a reading group of TCS students in and out of Yonsei University

◇ Organized seminar talks on various topics including mechanism design and quantum computing

Intern, Cornell University, USA Sept. 2022 – Dec. 2022

◇ Director: David B. Shmoys

Undergraduate intern, Yonsei University, South Korea Jan. 2017 – Feb. 2018

◇ Advisor: Hyung-Chan An

Undergraduate voluntary tutor, Yonsei University, South Korea

◇ CSI3108 Algorithm analysis Fall 2016, 2017

◇ CSI2103 Data structures Spring 2017

Web programmer, Republic of Korea Air Force, South Korea Nov. 2013 – Aug. 2015

◇ In fulfillment of mandatory military service